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College of Computing
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جامعة أم القرى
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Project Proposal:

Hand Gesture Control System for Touchless HCI Presentations

Section: 3

Group Members:

NAME	ID
Project Leader: Yosra Tamer	445013908
Batoul Hallaq	445013931
Afrah Ayob	445013880
Reham Khan	445013974
Razan Alturkistani	445002494

Dr. Hawazin Faiz Ali Badawi

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1. Problem Statement and Background

Human-Computer Interaction (HCI) has witnessed a rapid evolution, transitioning from traditional physical interfaces to Natural User Interfaces (NUI). In this context, gesture recognition technologies have emerged as a critical necessity in real-world scenarios that demand more fluid and liberated interaction. Rather than relying entirely on touching physical devices or remaining tethered to a control panel, these technologies enable users to manage digital systems remotely, thereby enhancing operational efficiency and reducing the physical constraints imposed by traditional tools such as the mouse and keyboard. This shift toward "touchless interaction" is not merely a technical luxury but a response to the growing need for more flexible and intelligent work and interaction environments.

Specifically within the educational environment, lecturers and students face significant challenges during demonstrations and presentations. Speakers are often forced to remain in close proximity to the computer to control slide transitions, which disrupts the instructional flow and reduces visual and kinetic communication with the audience. This project aims to address this issue by developing an advanced system based on Computer Vision and Machine Learning to provide full control over presentations via hand gestures alone. The system transcends simple page-turning to include precise interactive tasks, such as controlling the volume of displayed content. In Addition to a **"Virtual Pen"** feature that allows the lecturer to write or draw illustrative sketches over slides in real time, utilizes the hand as a smart Pointing tool, and provides control over educational "YouTube" media. This empowers the lecturer to play, pause, or navigate the video timeline with simple gestural movements from their position, ensuring a seamless and dynamic educational presentation.

Project Objectives and Technical Requirements:

- **Innovative Learning System Design:** Developing a model based on Machine Learning for gesture classification rather than fixed rules, ensuring high flexibility across different users.
- **Achieving Instantaneous Response:** Ensuring very low latency to enable tasks such as "writing during explanation" to feel natural and fluid.
- **Building a Custom Dataset:** Collecting and processing hand gesture data using the MediaPipe library to ensure accuracy under varying lighting conditions within classrooms.

2. Target User Group

The focus of this project is dedicated to serving the academic and educational community. The target groups include:

- **Faculty Members and Lecturers:** They represent the primary beneficiaries of the system, as it grants them the ability to manage lectures with complete freedom within the hall. They can control multimedia, adjust lighting, and record virtual notes while standing in front of the students, thereby elevating the quality of academic interaction.
- **Students and Researchers:** The project targets students across all academic levels, particularly during graduation project defenses and classroom presentations. The system helps them present their content in a professional and engaging manner, facilitating the explanation of concepts through touchless pointing and writing features.

3. Success Metrics and KPIs

The evaluation of the proposed system's success relies on three main pillars that ensure the model's **technical efficiency**, **temporal smoothness**, and **effectiveness in a real-world educational environment.** These pillars will be measured through a set of precise quantitative and qualitative indicators as follows:

- **First:** Mathematical and Classification Performance:

The system's ability to distinguish educational gestures with high accuracy will be evaluated using a Confusion Matrix to identify the degree of confusion between different movements. We will also rely on Precision and Recall metrics to ensure the system does not execute false commands resulting from a lecturer's random hand movements, in addition to the F1-Score to balance the two. A fundamental criterion here is measuring Robustness by maintaining an accuracy rate of no less than 90% across varying room lighting levels or varying distances between the lecturer and the camera.

- **Second:** Real-time Latency and Stability:

Since the system is used during live explanations, Latency is a critical indicator of success. The system must maintain a very low delay -not exceeding fractions of a second- to ensure real-time interaction. We will measure **Frames Per Second (FPS)**, targeting a processing speed that ensures high fluidity, particularly when using the "Virtual Writing" feature. Additionally, **"Tracking Stability"** will be measured to ensure there is no jitter or interruption in hand landmark detection as the lecturer moves within the field of view.

- **Third:** Functional Reliability and User Experience (UX):

The project aims to achieve a tangible superiority in field performance of no less than 70% compared to traditional methods in terms of task execution speed. We will evaluate the Task Completion Rate, measuring the lecturer's success in executing a sequence of consecutive commands (e.g., playing a video, then pausing it, then writing an illustration) without needing to return to the mouse or keyboard. Furthermore, the system's real-world reliability in handling "Complex Gestures" will be assessed to ensure it responds only to intended commands while ignoring natural hand gestures during speech.

4. Feasibility and Risk Assessment

The technical feasibility of this project is exceptionally high due to the reliance on advanced open-source software libraries such as MediaPipe and OpenCV, which have demonstrated high efficiency in tracking hand landmarks and processing them in real time using standard webcams without the need for expensive sensors.

From an applied perspective, integrating machine learning techniques grants the system flexibility in distinguishing gestures with an accuracy that surpasses traditional systems based on rigid rules.

The aspects of feasibility and risk assessment are summarized in the following points:

- **Technical and Operational Feasibility:** The system can convert simple movements into comprehensive system commands; specific gestures are allocated to control screen brightness or volume levels, and finger coordinates are utilized for digital drawing or writing directly over the presentation.
- **Environmental Risks:** System accuracy may be affected by low-light conditions in lecture halls or crowded backgrounds that might distract hand recognition algorithms. This issue can be resolved using various methods, such as hand signal normalization and relying on the MediaPipe library which is pre-trained on a massive, comprehensive, and diverse dataset for hand tracking and feature extraction.
- **Gesture Ambiguity:** There is a risk of confusion between different gestures. The team will work on effectively selecting gesture shapes to prevent such issues, in addition to training a specialized Deep learning model to accurately differentiate between movements to ensure no false commands are executed.
- **Latency Challenges:** To ensure a smooth experience for the lecturer, the temporal delay must not exceed fractions of a second. Focus will be placed on balancing the use of effective models with ensuring those models are lightweight to guarantee instantaneous movement processing.
- **Learning Curve:** New users may initially find difficulty in memorizing the gestures associated with each command. A user interface will be designed to illustrate the required gestures simply, in addition to providing the system with a User Manual to assist users in understanding the system and contribute to an easy and seamless user experience.

5. Ethical, Legal, and Constraints

Ethical and legal considerations are a top priority for this project. The "Hand Gesture Control for Tactile Interaction" system adheres to all seven principles of AI Ethics issued by the Saudi Data and AI Authority (SDAIA), implementing technical mechanisms that ensure governance, fairness, and transparency within the educational environment.

First: Project Commitment to SDAIA's AI Ethics Principles

1. Fairness
The project seeks to achieve digital justice by avoiding demographic and environmental biases.
Mitigating Under-representation Bias: The system relies on the MediaPipe library for feature extraction. This library is pre-trained on massive, comprehensive datasets, ensuring the system's immunity against bias regarding skin tones, hand sizes, or varying lighting conditions.

2. Privacy & Security
Privacy is the cornerstone of this system's design.
Local Processing: Video processing is conducted locally on the user's device without being stored or transmitted to external servers.
Data Abstraction: The system's role is limited to extracting hand landmark coordinates and converting them into digital commands, completely eliminating the risk of leaking facial images or personal identities.

3. Humanity
The system is designed to be an assistive tool that enhances human skills rather than replacing them.
The project aims to liberate lecturers from physical constraints (mouse and keyboard) to enhance eye contact and physical engagement with students, thereby supporting the pedagogical process and making classroom interaction more dynamic.

4. Social & Environmental Benefits
The project contributes to developing a smart and sustainable educational environment.
The system improves the efficiency of academic presentations and simplifies the explanation of complex concepts, ultimately raising the quality of national education.

5. Reliability & Safety
The project ensures high technical performance suitable for live use in classrooms.
The system targets a classification accuracy of at least 90% with very low latency to ensure smooth "virtual writing."
Mitigating Anchoring Bias: To ensure system stability throughout the duration of a gesture, Explainable AI (XAI) techniques will be used to analyze the temporal importance of frames. This ensures the model does not rely solely on the start of a movement while ignoring the rest.

6. Transparency & Explainability
The team is committed to making the system understandable and interpretable by users.
A user interface (UI) will be provided to clearly illustrate required gestures, accompanied by a comprehensive "User Manual."
Bias Analysis: A dedicated section in the project documentation (see Appendix on User-Centered Design and Bias Mitigation) details how the system handles confirmation, anchoring, and representation biases through Temporal Importance Maps and "blind user" testing to ensure a deep understanding of the system's decision-making mechanics.

7. Accountability & Responsibility
The system is subject to full human oversight with a clear distribution of responsibilities.
Task Distribution: Tasks are precisely distributed among team members to ensure the quality of every component.
Informed Consent: The system is committed to informing users (lecturers and students) that the camera is active for control purposes only, emphasizing that no conversations are recorded and no facial recognition is performed.

Second: Legal Considerations and Intellectual Property

The project is committed to the legal rights applicable within educational institutions:
Compliance with Intellectual Property (IP) Rights: Full adherence to the licensing and IP rights of the software libraries used and all open-source materials.
Alignment with Digital Privacy Policies: Ensuring system compatibility with the digital privacy regulations and policies implemented in educational institutions.

Third: Technical and Operational Constraints
Several technical and environmental factors influence the operational scope of the system:
Hardware Constraints: The success of the system depends on the quality of the webcam used; extremely low-resolution cameras may limit the system's ability to accurately recognize gestures.
Coverage Range and Positioning: The system’s functionality is limited to the distance and field of view covered by the camera. Tracking may be lost if the lecturer moves too far from the device or steps out of the visual range.
Software Integration: The system may face limitations when interacting with third-party applications that do not support programmed shortcuts or external command inputs.

6. User-Centered Design Approach to Enhance System Reliability and Mitigate Biases

Driven by our commitment to making this system a seamless tool that effectively serves the lecturer and supports the flow of the educational process, we have placed the "Human" at the heart of the design process. We are not merely seeking to build a technical algorithm; rather, we aim to create a User-Centric interactive experience that ensures fairness and transparency by providing solutions for the following expected biases:

1. Confirmation Bias
Definition: Seeking evidence that supports the current hypothesis while ignoring contradictory data.
This bias may surface during data collection; a developer might lean toward collecting samples of gestures they perform "perfectly," while ignoring cases where the hand is fatigued or in an uncomfortable position for a real user. Additionally, during testing, the developer might repeatedly try successful gestures to confirm that "the system works," failing to deeply analyze why other gestures fail.
Mitigation Strategy:
Blind Testing: Confirmation bias intensifies when developers test their own code because they know the system's "secrets." Therefore, we will involve users from outside the team to use the system without explaining the "ideal way" to perform a gesture, allowing us to objectively monitor system performance.

2. Anchoring Bias
Definition: Over-relying on the first piece of information received.
Since the model depends on a chronological sequence of video frames, it may "anchor" to the first five frames of a movement. If a motion begins in a way that resembles the "Virtual Pen" gesture, the model might persist with that classification even if the rest of the movement changes later, causing response lag or diagnostic errors.
Mitigation Strategy:
Temporal Saliency Maps: Utilize Explainable AI (XAI) methods to visualize which "time frames" were most influential in the decision. If it is discovered that the system focuses only on the initial frames, the LSTM model will be adjusted to assign greater weight to the end of the movement to break the initial "anchor."

3. Representativeness Bias
Definition: Occurring when the system makes a judgment because the input "resembles" a stored prototype, without considering other possibilities. For example, the system might classify a lecturer's random hand motion during an explanation as a command simply because it looked similar to a stored gesture.
Mitigation Strategy:
Visual Certainty Calibration: Instead of executing a command immediately upon detecting a similarity to a "prototype," we add a "Loading Circle" that appears around the lecturer's hand on the screen. The command is only triggered if the lecturer maintains the gesture until the circle is complete. Consequently, quick, random movements are insufficient to activate a command.
Negative Prototypes: Training the system on "common random movements" found in classrooms and classifying them as "ignored motions." This teaches the system that "similarity in form" does not always mean "similarity in meaning," a process known in research as misclassification error identification to avoid bias.

4. Global Representativeness Bias
Definition: This represents the fundamental challenge that combines selection bias, demographic bias, and background bias. It originates from a lack of diversity in the training database. In our project, this could lead to reduced efficiency or bias when used by different age groups, individuals with different skin tones, or under varying lighting and backgrounds.
Mitigation Strategy:
To address this challenge, our system relies on the MediaPipe library as a core component for Feature Extraction. Since this library is pre-trained by Google on massive, comprehensive datasets—including a vast diversity of hand sizes, skin tones, and lighting conditions—the system possesses an inherent immunity against traditional environmental and demographic biases.

7. Task Distribution

- Problem Statement and Background: Batoul
- Target User Group & Writing the Proposal: Reham
- Success Metrics and KPIs: Razan
- Feasaibility and Risk Assessment: Afrah
- Ethical, Legal, and constraints: Yosra
- User-Centered Design: Yosra

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